Chapter 15

15.1

a.
```
#include <stdio.h>

int main()
{
        int i = 1;
        int sum = 0;

        while(i < 11)
        {
             sum = sum + i;
             ++i;
        }
        printf("%d\n",sum);
}
```

b.
```
#include <stdio.h>

int main()
{
        int i;
        int sum = 0;

        for (i = 0; i <= 10; ++i)
            sum = sum + i;
        printf("%d\n",sum);
}
```

c.
```
#include <stdio.h>

int main()
{
        int i = 0;
        int sum = 0;

        while(i < 11)
             sum = sum + i++;
        printf("%d\n",sum);
}
```

d.
```
#include <stdio.h>

int main()
{
        int i = 0;
        int sum = 0;

        for(i = 0; i <= 10 ;)
                sum = sum + i++;
        printf("%d\n",sum);
}
```

15.3

```
#include <stdio.h>

main()
{
  int smallestNumber;
  int nextInput;

  scanf("%d",&nextInput);

  /* We need to set the initial value of smallestNumber to
     something other than 0 as in the original code */
  smallestNumber = nextInput;

  while (nextInput != -1) {
    if (nextInput < smallestNumber)
      smallestNumber = nextInput;
    scanf("%d",&nextInput);
  }

  if (smallestNumber != -1)
    printf("The smallest number is %d\n",smallestNumber);
  else
    printf("No numbers entered.\n");
}
```

15.5

a. (2,3) (2,4) (3,4) (2,5) (3,5) (4,5) (2,6) (3,6) (4,6) (5,6)

b. 22

c. The code can be made more efficient by changing the inner loop to:

```
       for (j = 2; j < (i/2); j++)
```

Doing so reduces the number of calls made to IsDivisibleBy by half.

15.7

The program, as is, allows someone to purchase a ticket without first making a reservation.

The following program accepts only 32 reservations for the 10 available seats. A reservation is required before a ticket can be purchased.

```
#include <stdio.h>

#define SEATS 10
#define MAX_RESERVATIONS 32

int main()
{
  int seatsAvailable = SEATS;
  char request;
  int number;
  int resStatus = 0;
  int resNumber = 0;


  do {
    scanf("%c",&request);

    if (request == 'R') {
      if (seatsAvailable && resNumber < MAX_RESERVATIONS) {
          printf("Reservation Approved\n");
          printf("Your reservation number is %d\n", resNumber);
          resNumber++;
      }
      else
        printf("Sorry, flight fully booked\n");
    }

    if (request == 'P') {
      printf("Enter reservation number to confirm purchase : ");
      scanf("%d", &number);

      if ((number >= resNumber) ||       /* Invalid number    */
          (resStatus & (1 << number)))   /* Already purchased */
        printf("Invalid reservation number.  Purchase denied.\n");
      else {
        resStatus = resStatus | (1 << number);
        seatsAvailable--;
        printf("Ticket Purchased!\n");
      }
    }

  } while (request != 'X');

 printf("Done! %d seats not sold\n", seatsAvailable);
}
```

Questions in the text denoted by the question mark icon:

Page 410: The problem with the code in Figure 15.3 is that the local variable **result** in function **AllSum** is uninitialized. It should be set to zero initially.